

UBER BLOCKCHAIN APP

A PROJECT REPORT

Submitted by

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*In fulfillment for the award of the degree
of*

BACHELOR OF ENGINEERING

in

Computer department



**LDRP Institute of Technology and Research,
Gandhinagar**

Kadi Sarva Vishwavidyalaya

January, 2024

LDRP INSTITUTE OF TECHNOLOGY AND RESEARCH GANDHINAGAR

CE-IT Department



CERTIFICATE

This is to certify that the Project Work entitled **“UBER BLOCKCHAIN APP”** has been carried out by **PATEL HET PARESHKUMAR (20BECE30149), PATEL JAINISH SHAILESHKUMAR (20BECE30153), PATEL JEEL DILIPBHAI(20BECE30156), PATEL JICAL HARESHKUMAR (20BECE30158)** under my guidance in fulfillment of the degree of Bachelor of Engineering in **COMPUTER DEPARTMENT** Semester-8 of Kadi Sarva Vishwavidyalaya University during the academic year 2023-2024.

Vaidehi Patel

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Presentation-I for Project-I

1. Name & Signature of Internal Guide	
2. Comments from Panel Members	
3. Name & Signature of Panel Members	

ACKNOWLEDGEMENT

We would like to express my profound gratitude to Prof. [Vaidehi Patel](#) as well as our HOD Dr. [Sandip Modha](#) who gave us the golden opportunity and the guidance to do this wonderful project, which also helped us in doing a lot of research and We came to know about so many new things We are thankful to them.

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ABSTRACT

In the world of cryptocurrencies, we have variety of different cryptos' like Bitcoin, Ethereum, Solana etc. in our wallet. Have you ever thought of paying for a cab using these cryptocurrencies? Today Our team come up with the solution with this problem. Our platform "UBER BLOCKCHAIN" provides Ethereum transaction for booking cab for a particular period. The technologies which we are used in this web app are SANITY, NEXTJS, REACTJS, ETHERJS, MAPBOX. The main aim for developing this web app is to make the use of crypto currencies in use for daily need of booking cab instead of regular currencies.

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1) INTRODUCTION

1.1) INTRODUCTION

1.2) SCOPE

1.3) PROJECT SUMMARY & PURPOSE

1.4) OVERVIEW OF THE PROJECT

1.5) PROBLEM DEFINITION

1 INTRODUCTION

1.1 INTRODUCTION

Crypto Currency are the future of currencies and the transaction of those currency will also be going to be high in demand due to no interference of any authorizing body and transparency with security.

However, an individual books a cab online by using regular digital cash nowadays. So why not to use Ethereum for booking a cab.

So, in this project we will try to build a UBER clone with little modification of Ethereum transaction where an individual will pay Ethereum instead of digital currency.

Here, we will learn new technologies like sanity for backend, ReactJs & NextJs for frontend and Meta mask & EtherJS for transactions in blockchain.

1.2 SCOPE

Day by Day increasing demand of crypto tokens like Solana, Bitcoin, Ethereum, Dogecoin, Polygon show wide scope of future transactions of these coins.

So, it becomes necessary to make use of these currencies in service sector also. Therefore an cab booking app like uber should be modified to accept cryptocurrency transactions.

Though no third-party dependency is needed in this transaction, makes the transaction more secure, transparent and reliable for naive user.

Transaction of famous crypto Ethereum in JULY 2020 is 1M approx., in JULY 2021 is 1.234M approx. and in JULY 2022 is 1.852M approx. Which shows booming growth in the transaction history of crypto tokens.

1.3 PROJECT SUMMARY AND PURPOSE

Cryptocurrencies are the future of money, and transactions involving them will be highly sought after because there will be no interference from any governing bodies and complete security.

In order to book a cab, we need digital currency or physical cash, why not to make use of the crypto currency for booking a cab. So we made a UBER clone DAPP (Decentralize app) so that Ethereum like crypto currency flow should be maintained in daily basis.

New technologies like Sanity for the backend, ReactJs and NextJs for the frontend, & Meta mask and EtherJS for blockchain transactions will all be covered here. Our app can be called as DAPP which means decentralized application which is based on blockchain technology where no one is the master.

1.4 OVERVIEW OF THE PROJECT

We are building famous UBER BLOCKCHAIN DAPP clone with

1. With a stunning design
2. Connected to Blockchain
3. Meta Mask Pairing
4. Sending and receiving the crypto tokens
5. Booking a cab
6. Map feature.

1.5 PROBLEM DEFINITION

The main idea to make this project is to learn how the blockchain technology works and transactions are done using popular technologies like meta mask, sanity, EtherJS.

We identified the problem that an individual is not able to use crypto Ethereum for making any type of payment in real world especially for booking a cab in our case. And the best part is it is not governed by any third-party bank. It is just like a normal uber web app but with hassle free Ethereum transaction feature to make use of crypto currencies.

2) LITERATURE REVIEW

2.1) TOOLS & TECHNOLOGY

2.2) HISTORY OF WORK DONE

2 TECHNOLOGY AND LITERATURE REVIEW

2.1 ABOUT TOOLS AND TECHNOLOGY

Technology used in this project are EtherJs, NextJS, ReactJs, SANITY.IO. We discuss about them in ensuing paragraphs.

1. NextJS & ReactJS

React (also known as React.js or ReactJS) is a free and open-source front-end JavaScript library for building user interfaces based on UI components. It is maintained by Meta (formerly Facebook) and a community of individual developers and companies.

Next.js is a flexible React framework that gives you building blocks to create fast web 3.0 applications.

2. Sanity.io

Sanity.io is the platform for structured content. With Sanity.io you can manage your text, images, and other media with APIs. You can also use the open-source single page application Sanity Studio to quickly set up an editing environment that you can customize. With Sanity.io you have access to a bunch of APIs, libraries, and tooling that helps you leverage the benefits of having all your content available as a single source of truth. This article will quickly walk you through some central concepts, giving you a head start.

Before we dive in, there are three important concepts to bear in mind:

- Sanity.io has a real-time datastore for structured content, and supporting APIs for assets, user management, and more.
- Sanity Studio is a user interface for managing content. It's an open-source React Single Page Application that you can customize and host wherever you want.
- There are also SDKs, libraries, and tools that let you query your content and integrate it with websites, services, and other applications; wherever you need content.

3. EtherJS

The ethers.js library aims to be a complete and compact library for interacting with the Ethereum Blockchain and its ecosystem. It was originally designed for use with ethers.io and has since expanded into a more general-purpose library.

Features

- Keep your private keys in your client, safe and sound
- Import and export JSON wallets (Geth, Parity and crowdsale)
- Import and export BIP 39 mnemonic phrases (12 word backup phrases) and HD Wallets (English, Italian, Japanese, Korean, Simplified Chinese, Traditional Chinese; more coming soon)
- Meta-classes create JavaScript objects from any contract ABI, including ABIv2 and Human-Readable ABI
- Connect to Ethereum nodes over JSON-RPC, INFURA, Etherscan, Alchemy, Cloudflare or MetaMask.
- ENS names are first-class citizens; they can be used anywhere an Ethereum addresses can be used
- Tiny (~88kb compressed; 284kb uncompressed)
- Complete functionality for all your Ethereum needs
- Extensive documentation
- Large collection of test cases which are maintained and added to
- Fully TypeScript ready, with definition files and full TypeScript source
- MIT License (including ALL dependencies); completely open source to do with as you please

2.2 BRIEF HISTORY OF WORK DONE

Our Project progress is divided into Eight phases of development which are Listed below

1 Setup Tailwind CSS

Firstly, we will initialize NextJs project and also add Tailwind CSS dependencies for the project.

2 Navbar

Then we will start our UBER clone app with making of nav bar.

Image of navbar

3 Add Map

Image of Map

Then after adding the navbar, we will setup the MAPBOX for showing map in the front end to the users for making navigation easier to the users.

4 Making Location selector and Ride Selector section

Then we are going to build location and ride selection section for the user where the user will select the location and book a ride for the journey.

Image of location selector

5 Authentication with Meta Mask

Then we will authenticate the user with using meta mask and proceed for transactions.

6 Calculate the Base Price and Make payment

At last, we will calculate the base price for the fair and then we will make payment using Meta Mask.

Image of proceeding the payment.

3) SYSTEM REQUIREMENTS

3.1) USER CHARACTERISTICS

3.2) HARDWARE & SOFTWARE

3.3) CONSTRAINTS

3.4) ASSUMPTIONS AND DEPENDENCIES

3 SYSTEM REQUIREMENTS STUDY

3.1 USER CHARACTERISTICS

User can transact the crypto using Meta Mask Extension after login and creating wallet into it.

User can select a location and book a cab.

User can see the location in the interactive & beautiful map UI.

User have access of transaction receipt for checking the transaction details in database.

User have wallet to store cryptocurrencies.

3.2 HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

Client:	Device with Internet Connection
Server:	Minimum 450 MHz processor Minimum 4 GB of Ram Internet Connection Super VGA Monitor (800*600)

Software Requirements:

Client:	Internet Connection
	Web Browser: Any HTML Compliant Browser Operating

3.3 CONSTRAINTS

3.3.1 Regulatory Policies

We use appropriate technical and organizational security measures to protect your stored personal data against manipulation, partial or complete loss, and unauthorized access by third parties. Our security measures are continuously being improved in line with technical developments.

Your Rights regarding your data

- 1 Right to confirmation
- 2 Right to access
- 3 Right to rectification
- 4 Right to erasure
- 5 Right to restriction of processing

3.3.2 Hardware Limitations

To get a full validated Ethereum node without storing +1TB of SSD, we must use Parity at the moment because it's the only client permitting pruning while is syncing the full blockchain

It makes an intensive use of random writes to disk. It's a lot included for a SSD

It makes an intensive use of syscalls which interrupt the CPU a lot

A SSD able to perform: 68 MB/s of random writes and 30.9 MB/s of randoms reads on average. +112GB of capacity (24/09/2018).

1. 13–14GB of RAM
2. A CPU able to handle a lot of interrupts

These requirements are not the minimums. With the above hardware, you will get a block synced per minute average of 340.2. You will eventually get synced with a rate of 4.29 and above but it will take more time

3.3.3 Interface to Other Applications

With the exception of the provider(s) of our websites, we do not make your personal data available to third parties unless you have expressly consented to it, if we are legally obligated to, or if this is necessary to enforce our rights concerning a contractual relationship.

Personal data collected via the Websites may be passed on to and/or accessed by the Website service provider(s). The Websites may be hosted on servers. The transfer of data is for the purpose of providing and maintaining the functionality of our websites.

3.4 ASSUMPTIONS AND DEPENDENCIES

1. Mass user adoption

The first line of defense for many critics is the argument that blockchain technologies can only replicate use cases that centralized systems already achieve, and that because of this, there is no reason for the average user to switch.

While I believe that during this first phase of adoption we will see these skeuomorphic use cases, I also think we haven't begun to understand yet what blockchain technologies can enable. The most powerful use cases for the internet didn't evolve until many years after initial adoption. However, we still have to prove adoption in the skeuomorphic period, and I think my hypothesis around adoption by investment addresses that.

2 User will have Tokens

We have assumed that the user will have sufficient amount of crypto token and then tries to do the valid transactions.

3 Dummy coins

We have created dummy coins which is our dependency for this project to show the transaction process between to parties through transaction ID.

4) STUDY ANALYSIS

4.1) CURRENT SYSTEM

4.2) PROBLEM IN CURRENT SYSTEM

4.3) SYSTEM REQUIREMENTS

4.4) FEASIBILITY STUDY

4.5) REQUIREMENTS VALIDATION

4.6) CLASS DIAGRAM

4.7) SYSTEM ACTIVITY

4.8) SEQUENCE DIAGRAM

4 SYSTEM ANALYSIS

4.1 STUDY OF CURRENT SYSTEM

Transaction in Ethereum is one of the core functions that play a role as an only thing that can change or update Ethereum states. It has a data structure and contains several elements that represent what activity happens in Ethereum blockchain network such as sender, recipient, date time, and amount of Gas. Here I will explain the inside of a transaction and elements to be explained are below.

Structure of Transaction

- Nonce
- Gas price
- Gas Limit
- Recipient (EOA, Contract Account)
- Value
- Data
- v, r, s(ECDsa)

4.2 PROBLEM AND WEAKNESSES OF CURRENT SYSTEM

- The project is still incomplete and is currently in the midst of a prodigious, multi-year upgrade that comes with substantial risk.
- Current scaling issues limit the ability of DAPPS and other projects to grow and create satisfying user experiences.
- Heavy competition from rival blockchains, such as Polkadot and Cardano, could chip away at Ethereum's adoption and popularity.

4.3 REQUIREMENTS OF NEW SYSTEM

4.3.1 User Requirements

- The following are the requirements that one user needs to take the leverage of this DAPP:
- The must have an account in Meta Mask.
- The user must be connected to the wallet to check his/her portfolio and to authenticate & being eligible for any transaction.
- The user must be connected to the internet.

4.3.2 System Requirements

Hard drive

- An Intel Pentium 4 processor or later that's SSE3 capable
- You'll need SSD storage to consistently handle necessary read/write speeds.
- Be sure to account for enough space on your drive until you run maintenance on your node.

CPU and RAM

- Check with client documentation to ensure the hardware you want to use is sufficient and supported.
- Resource usage can vary significantly between clients. Research the different clients if you're working with resource constraints.

Internet

- Ideally your internet connection should be reliable and as close to 24/7 as possible without interruption.
- Ensure your bandwidth can't be throttled and isn't capped so your node stays in sync and will be ready to validate when called.
- You need enough upload bandwidth too. As of May 2022, this is ~1.2-1.3 GB download and ~0.9-1 GB upload per hour, and is likely to increase.

4.4 FEASIBILITY STUDY

In a general perspective, this work investigates the feasibility of a blockchain-supported local energy market. Feasibility includes in this context a technological, economic and regulatory analysis. This thesis will especially focus on two objectives, where both covers unique parts of the total feasibility.

- How should trading mechanisms be designed as blockchain applications, and how much computational resources do they consume?
- How does a blockchain-supported local energy market comply with Norwegian regulations, and what is the economic costs related to implementation in a real system?

We answer these objectives by modelling a local energy market as blockchain applications, deploying them on a test blockchain, and analyses the performance. The market in question consists of three trading mechanisms, where all explores the benefits of flexible loads.

In addition, these three trading mechanisms are integrated with the same payment platform. In result, this could create an efficient market with little friction and a high utilization of flexible loads. Simulations are performed with base cases and sensitivity on both market structure and application design. The resulting computational resources are thereafter analyzed, which reveal important parts of the technical feasibility. Further, these technical results set a basis for the economic and regulatory discussion

4.5 REQUIREMENTS VALIDATION

The validation is most crucial part of any DAPP in our system the user gets validate using the meta mask wallet. Once the user gets connected to the internet and connect its wallet the web app starts working and able to do transactions.

4.6 CLASS DIAGRAM

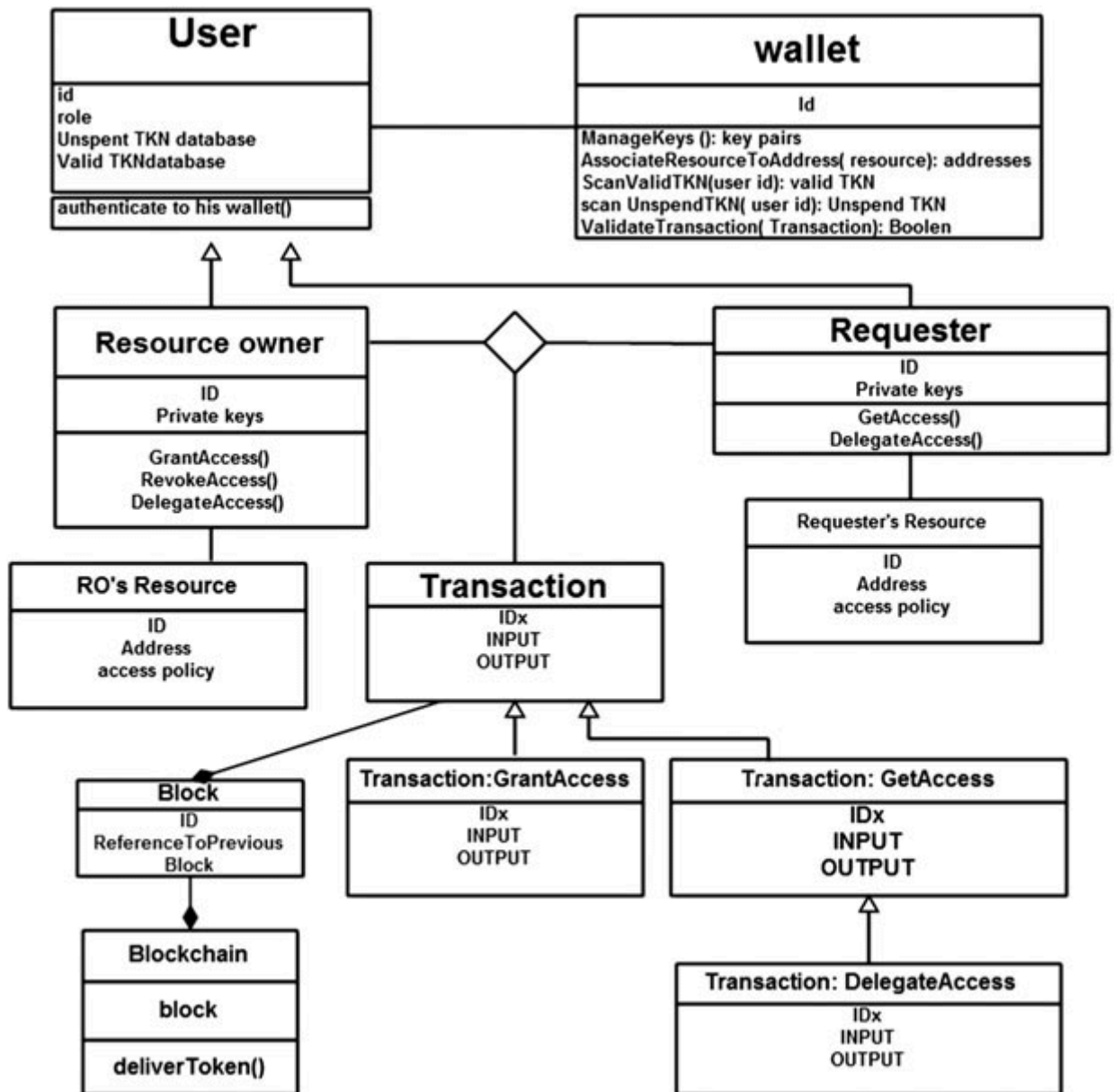


Figure: Class Diagram of crypto Transaction

4.7 SYSTEM ACTIVITY (USE CASE DIAGRAM)

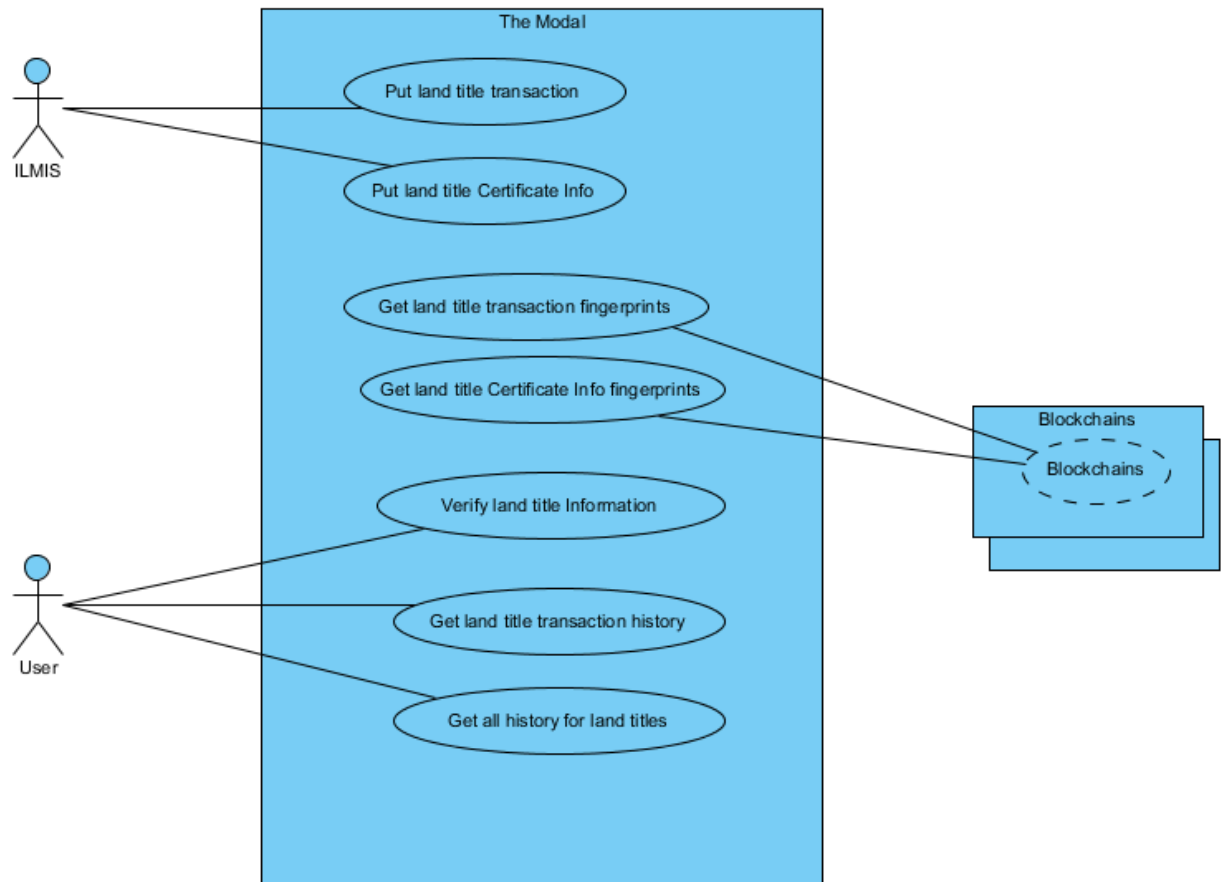


Figure: Use case Diagram for Crypto Transaction

4.8 SEQUENCE DIAGRAM

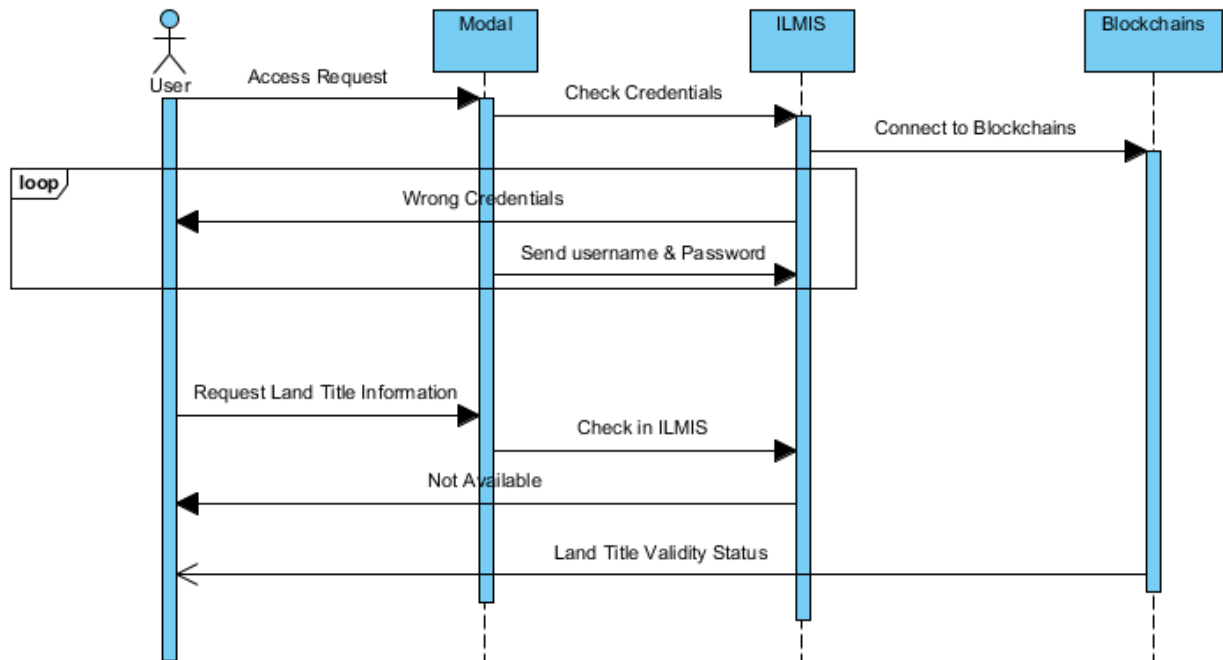
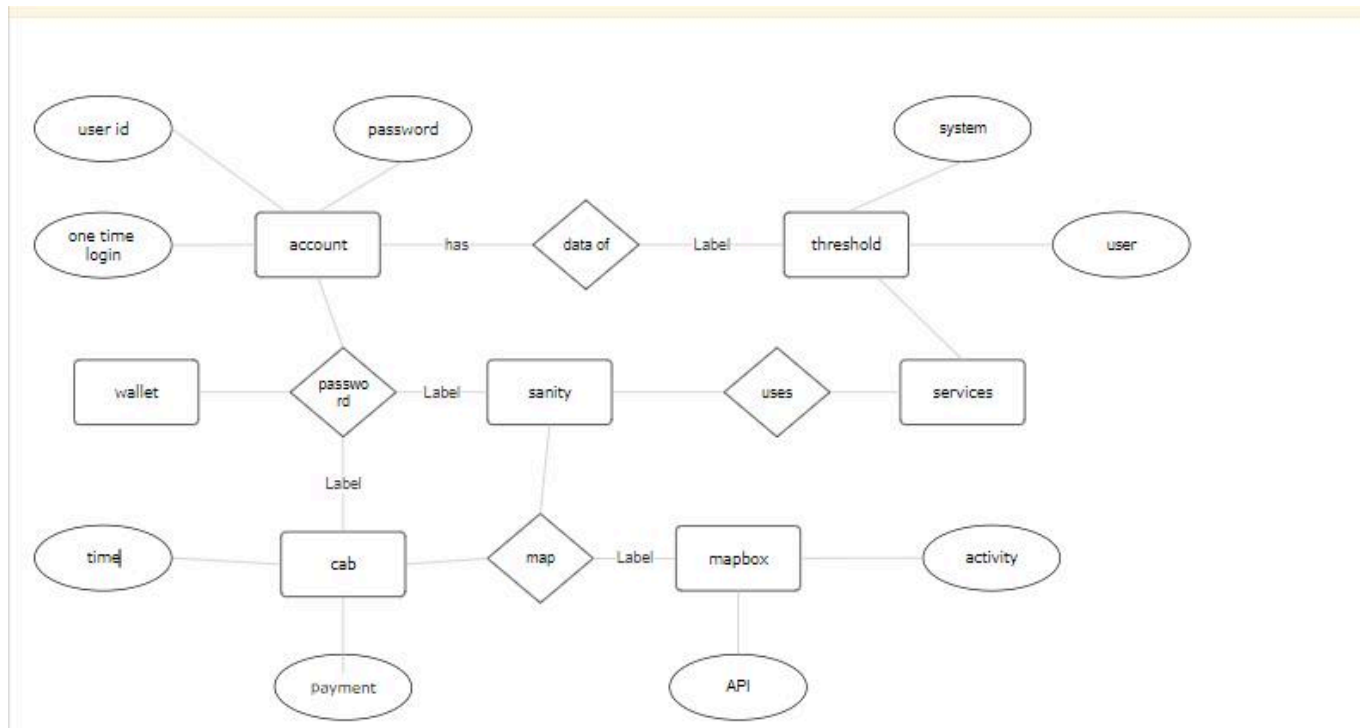


Figure: Sequence Diagram of crypto transaction

4.9) ER Diagram

5) SYSTEM TESTING

5.1) TEST CASES

5.SYSTEM TESTING

5.1TEST CASES

Test case ID	Test case Objective	Prerequisite	Steps	Input Data	Expected Output	Actual Output	Status
TC_1	Test Transaction module (without internet)	Transaction ID column should be enabled.	1.Enter amount to be paid. 2.Select currency. 3.Enter recipient wallet address.	1. 0.1 amount 2. Bitcoin 3. Wallet address	1.Connect to internet	1.Connect to internet	PASS
TC_2	Test Transaction module (internet)	Transaction ID column should be enabled.	1.Enter amount to be paid. 2.Select currency. 3.Enter recipient wallet address.	1. 0.1 amount 2. Bitcoin 3. Wallet address	1.Processing 2.Transaction complete	1.Processing 2.Transaction complete	PASS
TC_3	Test portfolio update	Without internet	Complete a transaction while using internet after that close the internet.	No input required.	Portfolio profile needed to be updated.	Portfolio not updated as it is not connected to the internet.	FAIL
TC_4	Test portfolio update	With internet	Complete a transaction while using internet.	No input required.	Portfolio profile needed to be updated.	Portfolio updated	PASS

LIST OF FIGURES

1 CLASS DIAGRAM

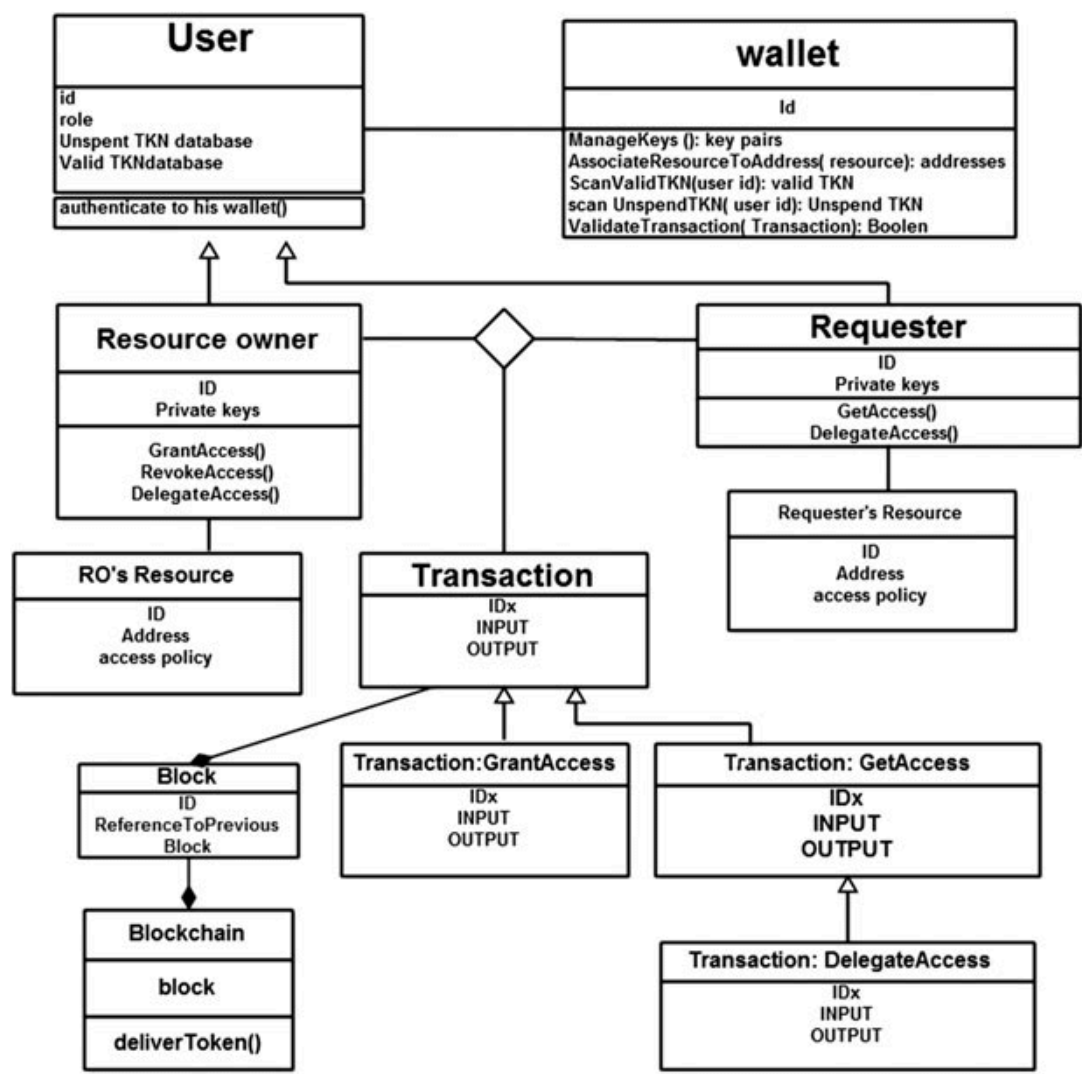


Figure: Use Case Diagram of Crypto Transaction

2 USE CASE DIAGRAM OF SYSTEM

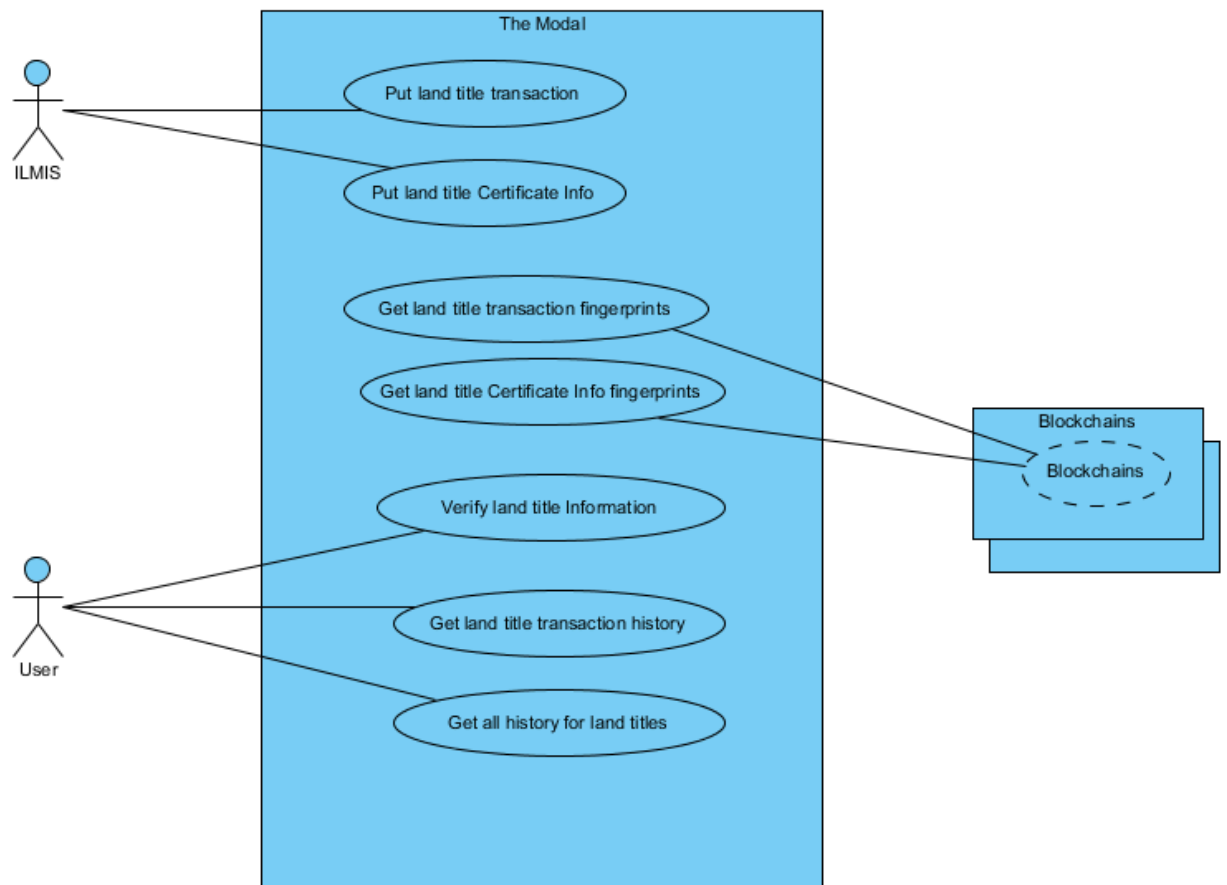


Figure: Use case Diagram of Ethereum model

3 SEQUENCE DIAGRAM OF ADMIN

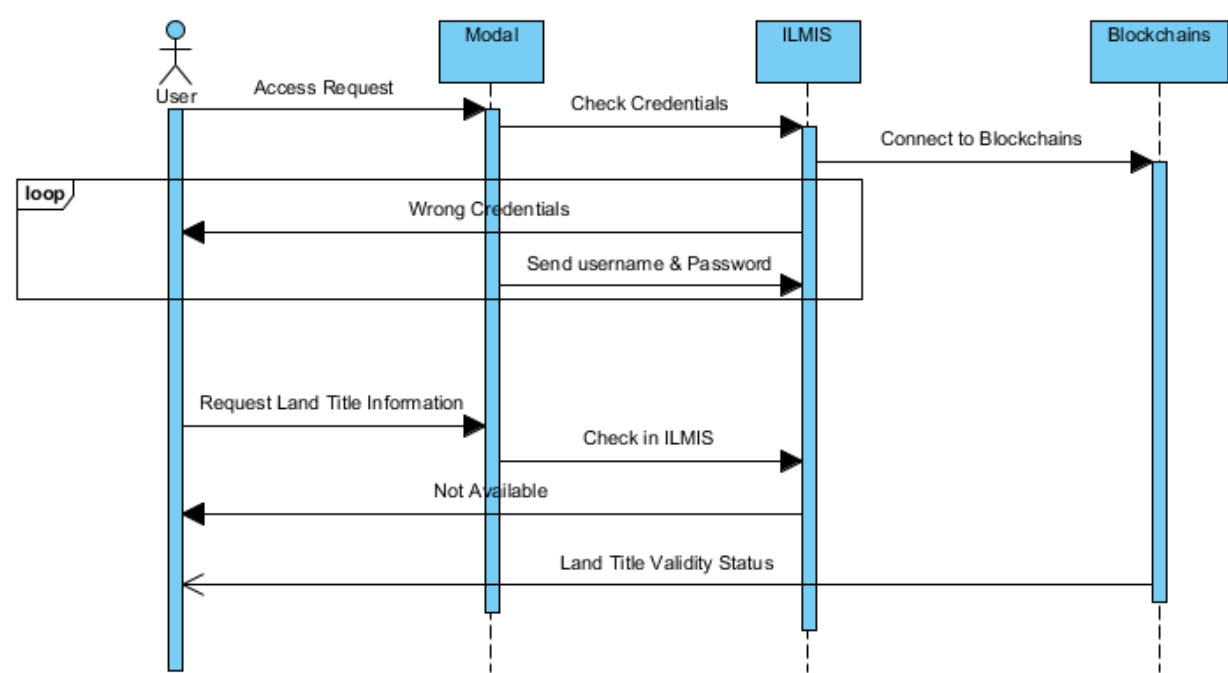


Figure: Sequence Diagram of system

LIST OF TABLES

Sanity Coinbase table.

Field	Data type	Constraints
Coins	Varchar (20)	Not null
Symbol	Varchar (20)	Not null
Contract address	HEX	Primary Key

6) CONCLUSION

6.1) CONCLUSION

6.0) CONCLUSION

Ethereum is a technology completely changing the way we transact. Thanks to smart contracts, the game-changing concept developed by one of Ethereum's co-founders, it is possible to replace existing standard contracts to improve transaction security, limit costs and potentially decentralize the world.

In terms of use and further development, Ethereum is becoming more mature, with a higher number of transactions than Bitcoin.

As the implementation of Ethereum 2.0 continues, we can safely assume that the network will grow and remain one of the biggest players in crypto.

Many believe this advancement will make it possible to have a sustainable and scalable protocol for future global use, so let's see if Ethereum 3.0 will live up to the expectations.

7) REFERENCES

7.1) REFERENCES

BIBLIOGRAPHY

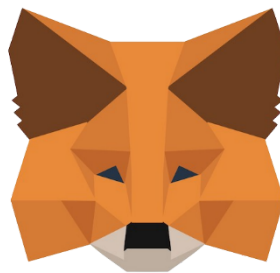


1. Third web official Page

Web 3.0 (Web3) is **the third generation of the evolution of web technologies**. The web, also known as the World Wide Web, is the foundational layer for how the internet is used, providing website and application services.

2. Sanity database

Sanity.io is **the platform for structured content**. It comes with an open-source editing environment called Sanity Studio that you can customize with JavaScript, and a real-time hosted data store.



3. Meta Mask

MetaMask is a cryptocurrency wallet that enables users to access the Web 3 ecosystem of decentralized applications (dapps).